

PALLAVIKA MANGALPALLY

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SUMMARY

AI Engineer with production software engineering experience building GenAI systems, RAG pipelines, and agentic AI workflows at enterprise scale. Uses Windsurf as a standard part of daily engineering workflow to accelerate development, debugging, and test authoring, alongside hands-on experience integrating LLM APIs (Claude, GPT-4, Gemini) into production code with management of token limits, latency, and cost tradeoffs. Skilled in hybrid search (dense and sparse retrieval), embedding lifecycle management, multimodal RAG, and building guardrails for hallucination mitigation using Python, LangChain, LangGraph, and GCP. Experienced defining KPIs and ROI metrics to evaluate AI-assisted workflow effectiveness, and translating customer use cases into scalable, full-stack GenAI capabilities that integrate with enterprise tools like Jira and Figma.

EDUCATION

Keshav Memorial Institute of Technology
Bachelor of Technology in Computer Science

Hyderabad
2020 – 2023

PROFESSIONAL EXPERIENCE

Virtusa Consulting Services Private Limited

Jan 2023 – Present

AI Engineer (Intern: Jan–May 2023 → Full-Time: Jan 2024–Present)

Hyderabad

- Completed GCP-Associate Cloud Engineer certification during internship; use Windsurf daily as a core part of engineering workflow for code generation, debugging, and code review
- Designed and implemented multimodal RAG pipelines integrating OCR and Vision Models for document re-extraction; architected vector indexing and retrieval strategies that reduced manual validation effort by ~70%
- Built agentic AI workflows with tool calling, planning, and orchestration patterns to ingest Jira stories; implemented prompt orchestration and context assembly achieving ~95% test case accuracy; reduced per-story test design time from 1–2 hours to under 5 minutes using AI-generated test cases
- Designed multi-agent orchestration system with memory and planning patterns to monitor Jira story changes and automate re-triggering of documentation workflows and approval emails
- Extracted requirements from PDFs and Figma using data ingestion and chunking strategies; modelled application screens as NetworkX Knowledge Graphs; implemented semantic search for test retrieval reducing manual test authoring by ~65% per sprint
- Built comprehensive RAG evaluation frameworks computing relevance, faithfulness, accuracy, and toxicity as KPIs; benchmarked RAG pipeline performance against industry alternatives using FastAPI/Django, and reported AI-assisted workflow ROI to stakeholders
- Implemented hybrid search combining dense (FAISS) and semantic retrieval over indexed code repositories to automate API test script generation from `.feature` files; integrated LLM APIs (Claude, GPT-4, Gemini) into these services with attention to token limits, latency, and cost tradeoffs
- Contributed to architecture decisions and technical design reviews for enterprise GenAI platform capabilities, including vector store infrastructure and retrieval optimization strategies; used Windsurf to build React and Node.js frontend interfaces for these tools, extending delivery across the full stack
- Owned end-to-end delivery of AI-assisted features across design, build, and production support within Agile sprint cycles, collaborating directly with client engineering teams and communicating project updates to ensure stakeholder alignment
- Contributed reusable RAG and agentic-orchestration components, prompt templates, and internal AI tooling standards to a shared knowledge base, accelerating onboarding and adoption of AI-assisted workflows across project teams

TECHNICAL SKILLS

- **Languages:** Python , SQL
- **AI-Assisted Engineering:** Windsurf, AI-Generated Testing, Prompt Engineering
- **AI / ML:** RAG Architecture, Multi-Agent Systems, Agentic AI, Guardrails & Hallucination Mitigation, Tool Calling & Agentic Patterns, NLP, Conversational RAG Bot
- **LLMs & Orchestration:** Claude (Anthropic), GPT-4, Gemini Pro, ADK, LangChain, LangGraph, Token/Latency/Cost Management
- **Vector DBs & Retrieval:** FAISS, Pinecone, ChromaDB, Vector Search, Hybrid Search (Dense + Sparse), Embedding Lifecycle Management, Chunking Strategies
- **ML Frameworks:** Hugging Face Transformers
- **Compliance & Governance:** Data Governance, PII Handling, Access Control, SOC2 & GDPR Awareness, Secure AI Deployment Practices
- **Backend & APIs:** Django, FastAPI, REST APIs, Figma API, SharePoint, Gherkin/BDD, Multimodal RAG, JSON Parsing
- **Frontend/Full-Stack:** HTML, CSS, JavaScript
- **Cloud & DevOps:** GCP, AWS (Basics), GitLab, GitHub Actions, CI/CD, Docker
- **Data & Databases(SQL or NoSQL):** MongoDB, SQLite, Pandas, NumPy
- **Tools & Frameworks:** OCR & Vision Models, NetworkX Knowledge Graphs

CERTIFICATIONS & ACHIEVEMENTS

- Google Cloud Certified – Associate Cloud Engineer | Virtusa Certified – GenAI Assisted Engineer
- Agile 101 Certification | Cyber Security Certification – Verzeo
- **Top 5 Finalist** – Virtusa Women's Hackathon (Innovation & Problem Solving)